The Data Guy

The Data Analyst Interview Pack

The questions that decide fresher interviews, answered step by step with visuals. SQL, Excel, the project story, resumes and the job hunt.

SQL · EXCEL · PROJECT STORY · RESUME · JOB HUNT

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7 years in data · Free edition · November 2026

INSIDE THIS PACK

How to use this pack

Don't read it once and close it. Each page is one question you will face. Read the page, then answer it **out loud** without looking. That's the only way it sticks in an interview.

1. The project story question (the one that decides most offers)
2. SQL: the Nth highest salary, and its 3 follow-ups
3. SQL: RANK vs DENSE_RANK vs ROW_NUMBER
4. SQL: top earner in every department (PARTITION BY)
5. Excel: the 5 questions behind "Advanced Excel"
6. Excel: VLOOKUP vs XLOOKUP
7. Excel: month-on-month growth in 30 seconds
8. The case question: "Sales dropped 15%. Why?"
9. Resume: what a recruiter checks in 6 seconds
10. Resume: the one-line bullet formula
11. Job hunt: the 3-2-1 rule + a referral message
12. Your readiness checklist

Tip: Do one page a day. In 12 days you will have answered every question here at least once, out loud.

01 · PROJECT STORY

"Walk me through a project you're proud of."

It sounds harmless. It's the question that decides most offers, because the interviewer is not grading your project. They're grading whether you can be trusted with a business problem.

× **Weak:** "I built a Power BI dashboard showing sales trends by region, with filters for date and product."

✓ **Strong:** "One region's sales were flat for two quarters and nobody knew why. I combined sales and campaign data, found the drop matched a territory change, and built a dashboard the sales lead could use himself. They reassigned two accounts."

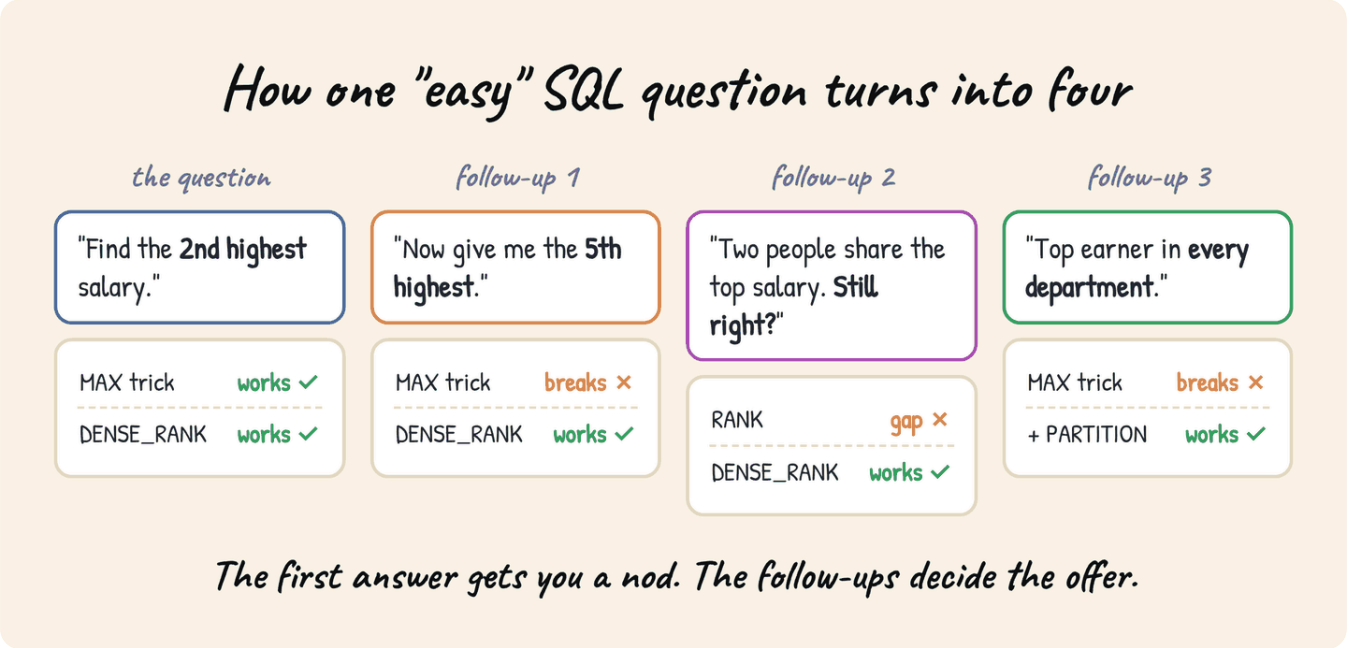
The 4-part structure behind every strong answer:

1. **Problem:** what was broken, in plain business language.
2. **Approach:** what you chose to do, and one decision someone else might not have made.
3. **Obstacle:** something that didn't go smoothly. Friction makes stories believable.
4. **Outcome:** a number, or a decision someone made because of your work.

No work experience? Frame a course or Kaggle project around a business problem with a stakeholder in mind. The structure matters more than whether it was a "real" job.

"Find the 2nd highest salary."

The first answer gets a nod. The follow-ups decide the round.




```sql
SELECT salary
FROM (
  SELECT salary,
         DENSE_RANK() OVER (ORDER BY salary DESC) AS rnk
  FROM employees
) t
WHERE rnk = 2;   -- change 2 to any N
```

Say this: "I'd use DENSE_RANK so ties are handled and I can change N without rewriting the query."

03 · SQL

RANK vs DENSE_RANK vs ROW_NUMBER

All three number your rows. They only disagree on ties, and interview data always has a tie.

Same data. Three functions. Three different answers.

Employee	Salary	ROW_NUMBER	RANK	DENSE_RANK
Asha	95,000	1	1	1
Ravi	95,000	2	1	1
Meera	80,000	3	3	2 ✓
Karan	72,000	4	4	3
Neha	72,000	5	4	3
Arjun	60,000	6	6	4

ROW_NUMBER
Every row gets a new number. Ties are split at random.

RANK
Ties share a number, then it **skips**. There is no rank 2 at all.

DENSE_RANK
Ties share a number, **no gaps**. Rank 2 = the real 2nd highest.

"2nd highest salary" = WHERE dense_rank = 2

- **DENSE_RANK:** any "Nth highest" or "top N values" question.
- **RANK:** competition-style ranking, where a tie for 1st means nobody is 2nd.
- **ROW_NUMBER:** exactly one row per group, like removing duplicates.

The trap: with RANK, `WHERE rnk = 2` can return zero rows. No error. Just silence.

04 · SQL

Top earner in every department

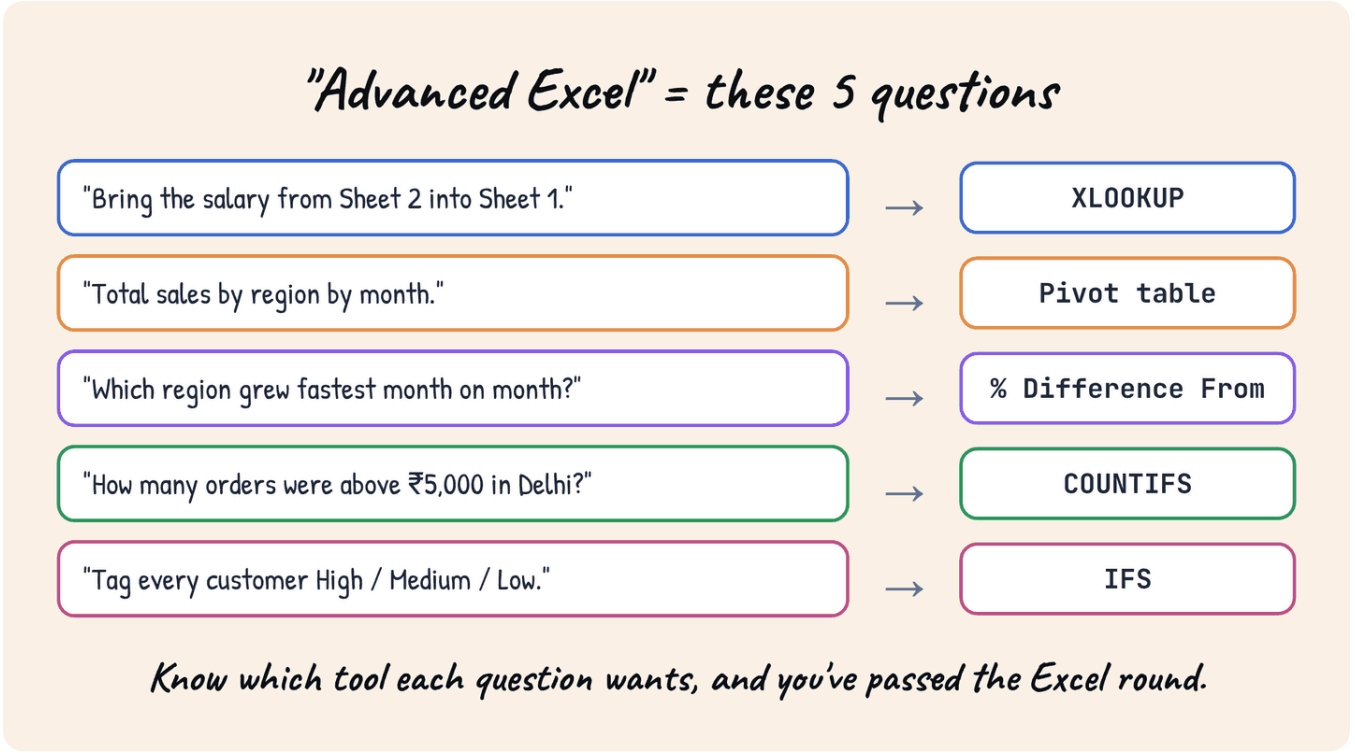
One keyword: `PARTITION BY`. It restarts the ranking inside every group.




```
SELECT department, name, salary
FROM (
  SELECT department, name, salary,
         DENSE_RANK() OVER (PARTITION BY department
                            ORDER BY salary DESC) AS rnk
  FROM employees
) t
WHERE rnk = 1;
```

Same pattern answers: top N per group, latest order per customer, best-selling product per category.

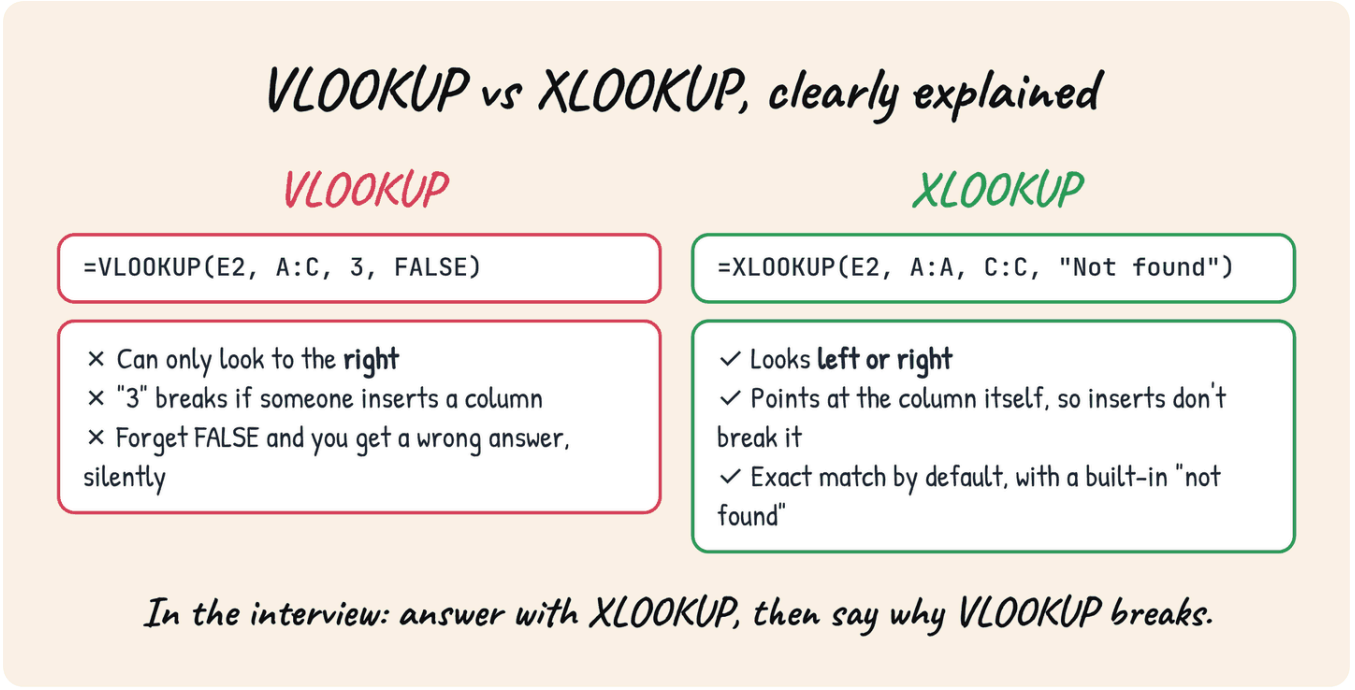
The 5 questions behind "Advanced Excel"




```
Q4  =COUNTIFS(City,"Delhi", Amount,">5000")
Q5  =IFS(B2≥50000,"High", B2≥20000,"Medium", TRUE,"Low")
```

Rule: interviewers watch which tool you reach for. The right tool takes 30 seconds. The wrong one runs out the clock.

VLOOKUP vs XLOOKUP



Say this: "I'd use XLOOKUP because it doesn't break when columns move and it can look left. On older Excel versions I'd use INDEX-MATCH, which works the same way."

Why it lands: that last sentence shows you've worked with real company files.

07 · EXCEL

Month-on-month growth in 30 seconds



The same **Show Values As** menu also gives you % of total, running total and rank. No helper columns, no formulas.

Practice: download any sales dataset from Kaggle and find the fastest-growing region in under a minute.

08 · CASE ROUND

"Sales dropped 15% last week. Why?"

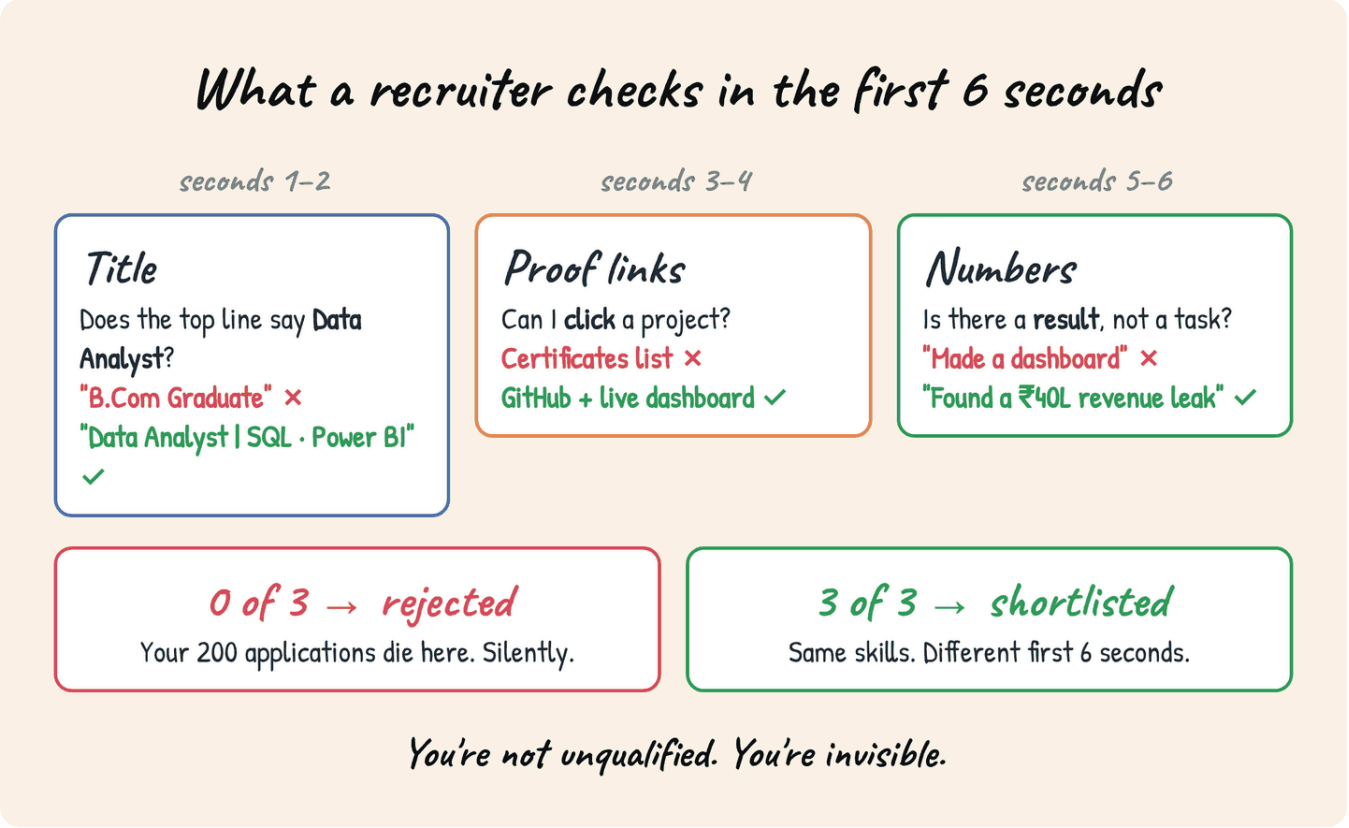
Don't guess a reason. Interviewers are testing your structure, not your luck.

1. **Check the data.** Is the drop real, or a tracking or reporting issue?
2. **Check the calendar.** Holiday, festival, month-end, payday shift?
3. **Segment the drop.** Region, product, channel, device, new vs returning customers. Find where it happened.
4. **Form a hypothesis.** "Most of the drop is mobile in the South, starting Tuesday."
5. **Verify and recommend.** What data would confirm it, and what would you do next?

Say this to start: "Before guessing, I'd first confirm the data is correct, then split the drop by region, product and channel to see where it's coming from."

Bonus points: ask a clarifying question first. "Is this compared to last week or the same week last year?"

What a recruiter checks in 6 seconds



- **Title:** write "Data Analyst | SQL · Excel · Power BI", even as a fresher.
- **Proof links:** GitHub and a live dashboard in the top third of the page.
- **Numbers:** every project bullet ends in a result, not a task.

The one-line bullet formula

Same project. One gets ignored. One gets a call.

before

- Made a sales dashboard in Power BI using a Kaggle dataset.

↓ *add the 3 missing parts* ↓

Action verb	Scale	Insight
Built, found, cut	₹4.2 Cr · 5 regions	what you discovered

after

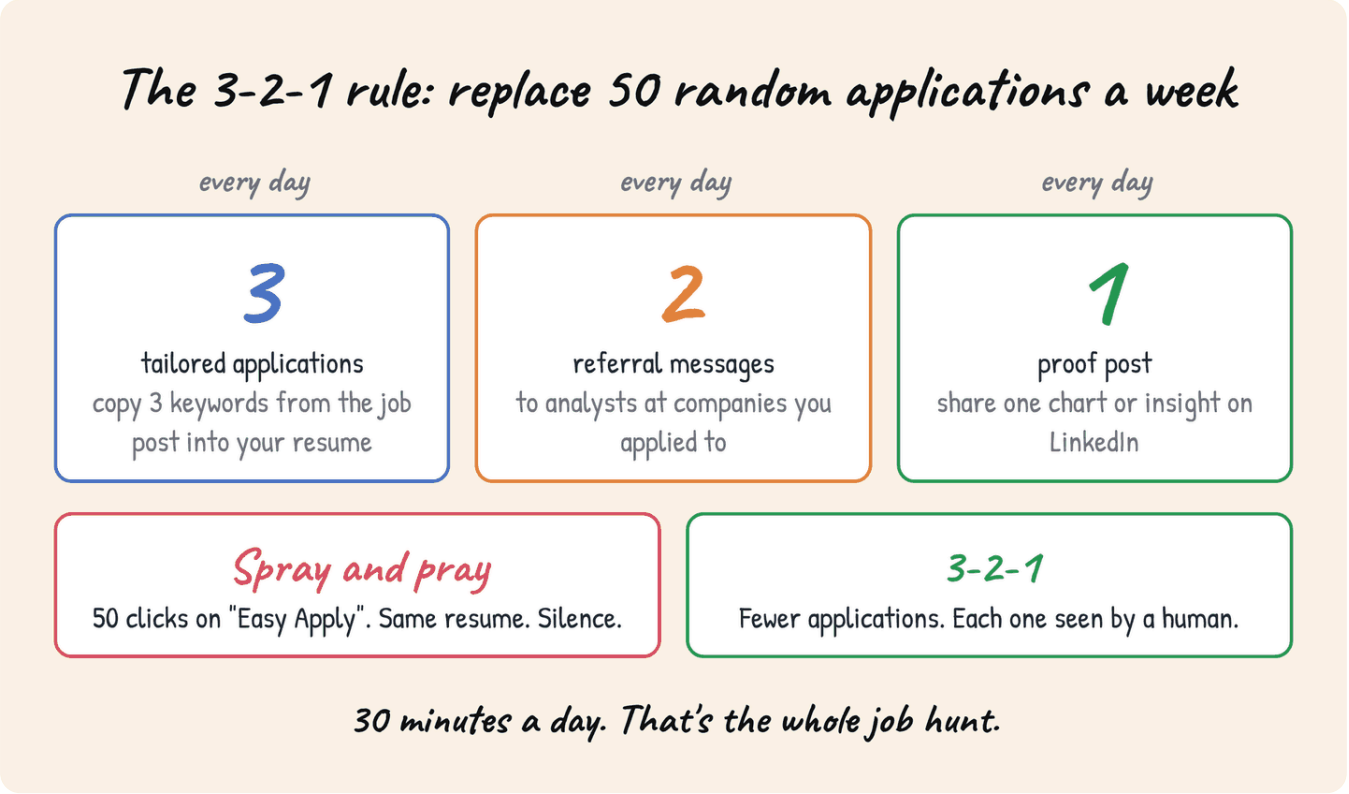
- **Built** a Power BI sales dashboard tracking **₹4.2 Cr revenue across 5 regions**; **found** one region drove 38% of the Q3 drop and flagged it with a live link.

Recruiters don't hire tasks. They hire findings.

```
[Verb] a [tool] [thing] tracking/covering [scale];
found [insight] and [what you did with it].
```

Do it now: rewrite your top 3 resume bullets with this formula. It takes 10 minutes.

The 3-2-1 rule



Referral message to copy (LinkedIn):

```
Hi [Name], I just applied for the Data Analyst role at [Company].
I built a [one-line project + result], here's the link: [link].
If you think I'm a fit, would you be open to referring me?
Either way, thanks for reading this.
```

12 · CHECKLIST

You're interview-ready when you can:

- ☐ Tell your best project story in under 90 seconds, using the 4 parts
- ☐ Write the Nth-highest query from memory, and explain why DENSE_RANK
- ☐ Explain RANK vs DENSE_RANK vs ROW_NUMBER with an example
- ☐ Solve "top N per group" with PARTITION BY
- ☐ Answer all 5 Excel questions without Googling
- ☐ Show month-on-month growth in a pivot in under a minute
- ☐ Walk through "sales dropped 15%" with a structure
- ☐ Point to 2 resume bullets that end in a result
- ☐ Send 2 referral messages a day

Want the full path, not just the questions?
The 90-Day Data Analyst System takes you from Excel to SQL to Power BI to Python, day by day: 27 projects, 100+ interview questions with model answers, resume and LinkedIn templates, and me as your mentor for a full year.

See what's inside →

Questions? Reply to any of my emails, or find me on Instagram **@thedataguy16**.